2.5 Viète's Product

WORKSHEET

The question

You may assume that all infinite sums and products in this question converge.

(i) Prove by induction that for all positive integers n,

$$\sinh x = 2^n \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)\sinh\left(\frac{x}{2^n}\right)$$

and deduce that, for $x \neq 0$,

$$\frac{\sinh x}{x}\cdot\frac{\frac{x}{2^n}}{\sinh\left(\frac{x}{2^n}\right)} = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right).$$

(ii) You are given that the Maclaurin series for $\sinh x$ is

$$\sinh x = \sum_{r=0}^{\infty}\frac{x^{2r+1}}{(2r+1)!}.$$

Use this result to show that, as y tends to 0, $\dfrac{y}{\sinh y}$ tends to 1.

Deduce that, for $x \neq 0$,

$$\frac{\sinh x}{x} = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)\cdots.$$

(iii) Let $x = \ln 2$. Evaluate $\cosh\left(\frac{x}{2}\right)$ and show that

$$\cosh\left(\frac{x}{4}\right) = \frac{1+2^{\frac{1}{2}}}{2\times 2^{\frac{1}{4}}}.$$

Use part (ii) to show that

$$\frac{1}{\ln 2} = \frac{1+2^{\frac{1}{2}}}{2}\times\frac{1+2^{\frac{1}{4}}}{2}\times\frac{1+2^{\frac{1}{8}}}{2}\cdots.$$

(iv) Show that

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2}\times\frac{\sqrt{2+\sqrt{2}}}{2}\times\frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2}\cdots.$$

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What is the double-angle formula for sinh? (If you are not sure, write $\sinh 2y$ and $\sinh y \cosh y$ in terms of exponentials and compare.)

STAGE 1 Halving repeatedly

Ask yourself Suppose the formula holds for n. The last factor is $\sinh(x/2^n)$. How can you rewrite it to get the formula for $n+1$?

STAGE 3 A formula for $\ln 2$

Ask yourself What is the general pattern for $\cosh(x/2^k)$? What happens when you multiply all the denominators $2 \times 2^{1/2^k}$ together?

STAGE 4 Viète's formula

Ask yourself Part (iv) has π in it, so it must come from ordinary trigonometry, not hyperbolic functions. Does everything in parts (i) and (ii) still work with sin and cos in place of sinh and cosh?

Pause and try Choose x so that the left-hand side is $2/\pi$, and work out the first few factors. The half-angle formula $\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}$ (for $0 \le \theta \le \pi$) will be useful.

Try this next

1 **The 1990 version.** (From STEP I, 1990.) Show directly, using only the double-angle formula for sin, that

$$\cos\left(\frac{\alpha}{2}\right)\cos\left(\frac{\alpha}{4}\right)\cdots\cos\left(\frac{\alpha}{2^n}\right) = \frac{\sin\alpha}{2^n\sin(\alpha/2^n)},$$

and deduce Viète's formula in the form $\dfrac{\pi}{2} = \dfrac{1}{\sqrt{\frac{1}{2}}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdots}$.

2 **Another angle.** Use $x = \frac{\pi}{3}$ in the product for $\frac{\sin x}{x}$ to write $\frac{3\sqrt{3}}{2\pi}$ as an infinite product, and simplify the first two factors.

3 **How many factors?** Use a calculator to find how many factors of Viète's product are needed to give π correct to 4 decimal places. Explain why each extra factor gains about 0.6 of a digit.

4 **A formula for $\ln 3$.** Repeat part (iii) with $x = \ln 3$ to write $\frac{1}{\ln 3}$ as an infinite product. How many factors are needed to give $\frac{1}{\ln 3}$ to two decimal places? Why does this product converge more slowly than Viète's?

Further reading

- **Petr Beckmann, *A History of π*** (St Martin's Press). Archimedes, Viète, Wallis and the rest, told with gusto.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). The serious mathematics of computing π, including the method of Section 2.6. Chapter 1 is accessible with this chapter behind you.
- **Eli Maor, *Trigonometric Delights*** (Princeton University Press). Contains a chapter on Viète and his product.

The full guided solution, and what the question leads to at university, is in Section 2.5:
https://beyondstep.keithmcnulty.org/ch02/viete.html